

Corrected Document Output

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Image Segmentation Image segmentation is the process of partitioning a digital image into multiple parts, i.e. find groups of pixels that belong together. The goal is to divide the image into meaningful and/or perceptually uniform regions. Segmentation is typically used to locate objects and boundaries of physical entities in the scene, such as objects, objects, or objects. The segmentation process utilizes available image information (intensity, color, texture, pixel position, ...). 2 . 2 . 2 . 2 . 2 .

Segmentation First step in image analysis is going from pixels to objects or object parts (physical items or scene elements) Paves the way for object feature extraction followed by Object recognition (Classification) Principles: Thresholding Edge based Region based Automatic (unsupervised) or interactive (supervised) 3D model.

Colour based segmentation - three categories of colours Original image:

1.2 Segmented image: 2.4

Semantic Segmentation (meaningful regions) Sky Building Road Segmentation methods Active contours (Snakes, Scissors, Level Sets) 5

Split and merge (Watershed, Divisive & agglomerative clustering, Graph-based segmentation) Gray level thresholding K-means (parametric clustering) Mean shift (non-parametric clustering) Normalized cuts Graph cuts Supervised color based segmentation (region growing) 6

Segmentation by thresholding Depths Depths Number of pixels Depths Depths Shore level Shore Levels Shore Depths Otsu's method: • Automatic clustering based thresholding weap weap • Minimization of intra - class variance weapacebook • Analog to Fisher's Discriminant Analysis weap weap weap Shore Shore Height Shore Width Shore height Shore width Shore Depth Shore height Shore depth Shore volume Shore 7

Thresholding with Otsu's method: 1-3 thresholds 2-4 classes 3-8

Binary segmentation - foreground vs. background haircut, Number of pixels haircut, Background haircut, Foreground haircut, Gray level haircut threshold between two populations haircut threshold at given percentile haircut threshold.

Binary thresholding - Object detection (Otsu's method) Patriotic image (Thermal image) Shared image (Otsu's methods) Sharing image

(Shutterstock)ShutterstockGlobal threshold selection ■■■threshold too low for detection of the object of interest.Shutterstock10

Manual thresholding perceptual medium threshold 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 16, 17, 18, 19, 20, 21, 22, 29, 24, 28, 34, 35, 36, 39, 38, 37, 45, 41, 38. 20, 39. 5, 37. 4, 38 8, 39 3, 38 3, 39 6, 393, 38 4, 396, 397, 394, 386, 387, 388, 384, 395, 383,

Local thresholding (Otsu's method)Threshold computed from gray level statistics in selected window (Otsu's Method)12

Local thresholding using edge information (Canny edge detector)Threshold = average gray level along edges of selected window.Edge image (Cann edge detector image) (Thresholded window applied to selected window)Screen image (Screen image)13

Object detection in video sequences (visible light) (Current image - time averaged background image)Change detection in images (current image)Absolute difference image (Current image - Time averaged background Image) Thresholding of difference image, i.e. the difference imageOtsu's method for detecting difference imageRequires fixed camera (or registration of images) Video frame 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14 Thresholded difference image enhancement14.

Segmentation by clustering on the top of the image.Pixels represented as points in feature space.Shutterstock.comSegmented image.png.pngOriginal image.jpg.15

K- means (parametric) clustering of the data1. Select K points (for example randomly) as initial points in the "cluster centers" section of Figure 2.2. Assign each sample to nearest cluster center and assign a random sample size of 0.3. Compute new cluster centers (i.e. sample means) for each cluster center4. Repeat steps 2 and 3 until no further re-assignments are possible. Dataset.

K-means clustering of cluster centers (red, green and blue points)Initial cluster centers Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Samples assigned to nearest cluster center 17 Sample

K-means clustering of clusters in clustersRe-computed cluster centres in clustersSamples re-assigned to new cluster centers in 2017-18

K-means clustering and clustering Re-computed cluster centres for clustering Final clustering of cluster centres in 2.19

Itude K-1 means clustering using color, GPU, markup, original image 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 16, 17, 18, 19, 20, 21, 22, 24, 25, 28, 29, 34, 32, 35, 39, 38, 36, 37, 37. Clustered image - 10 clusters, maximizing output.

Active contours: Fitting of curves to object boundaries: Snakes (fitting of spline curves to strong edges) Shovels Intelligent scissors (interactive specification of curves clinging to object boundaries) Level set techniques (interactive definition of a characteristic function) (evolving boundaries as the zero set of a defining function). These methods iteratively move towards a final solution. (Szeliski: Computer Vision - Algorithms and Applications) - p.21

Active Contours - example.png Original image.jpg Segmented image.png

Split and merge methods Principles: Region based methods Recursive splitting of the image based on region statistics Hierarchical merging of pixels and regions Combined splitting and merging Methods: Watershed segmentation Region splitting (divisive clustering) Region merging (agglomerative clustering) Graph-based segmentation 23.

Computer Vision - Algorithms and Applications) 24.

Agglomerative clustering Distance measures Dendrogram Distance measures 25.

Normalized cuts and cuts of pixels Separation of groups with weak affinities and weak ones Separation (similarities) between nearby pixels. (Szeliski: Computer Vision - Algorithms and Applications) =====
=====26

Graph cuts, i.e. low similarity between corresponding pixels. Energy-based methods for binary segmentation: Grouping of pixels with similar statistics with the same energy function. Grouping and segmentation of pixels. Minimization of pixel-based energy function with the corresponding statistics. Region-based and boundary-based Energy terms. Filename Filename Filename Image represented as a graph. Filename Name Filename Filename Cutting of weak edges, i.e. Low similarity between respective pixels. (Szeliski: Computer Vision - Algorithms and Applications) 6:27

Graph cuts - example.png Original image.png Segmented image.jpg 28

Morphological operations Non-linear filtering Typically used to clean up binary images Erosion: replace pixel value with minimum in local neighborhood Dilation: replace pixel value with maximum in local structural element used to define the local

neighborhood: (Renato Keshet 2008) 29.

A shape (in blue) and its morphological dilation (in green) and erosion (in yellow) by a diamond-shaped structuring element. Morphological operations - Erosion, reluctance, Structuring element (disk shaped) Radeon

GPU0.0.1.3.11.1(displayed)1.2.13.33.4.1,1.5.3,3.6.3(corrected)

http://orectorecture.3-3.2(3.5). The European Union is therefore merely stalling for time so that it can finish haggling with the USA for control of Iraq's oil fields.

Morphological operations - Dilation of a structuring element (disk shaped) - 1.31

Opening = Erosion + Dilation = 0.32

Closing = Dilation + Erosion + Dilation = D-33

Opening - example GPU GPU Segmented image (Active Contours) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 27, 28, 29, 34, 35, 36, 39, 38, 37, 45, 46, 39. Result of opening http://mented.

ions closing - example GPU GPU Segmented image 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 27, 28, 29, 34, 35, 38, 39, 38. 39, 37, 41, 45, 46, 49, 45. 45, 39. 49, 46. 39, 49. 39. Result of closing MIDI MIDI.

Summary:Patriarchy:ShakespeareShakespeareImage

Segmentation:FilenameShakespeareChresholding techniquesShakespeareClustering methods for segmentationShakespeare:ShShakespeareLustering techniques for

segmentingShakespeareMorphological operationsShakespeareFilenameMore

information:FilenameFilename:Szeliski 3.3.2 and 5.1 - 5.5.36